

Session 1 – Project Initialization

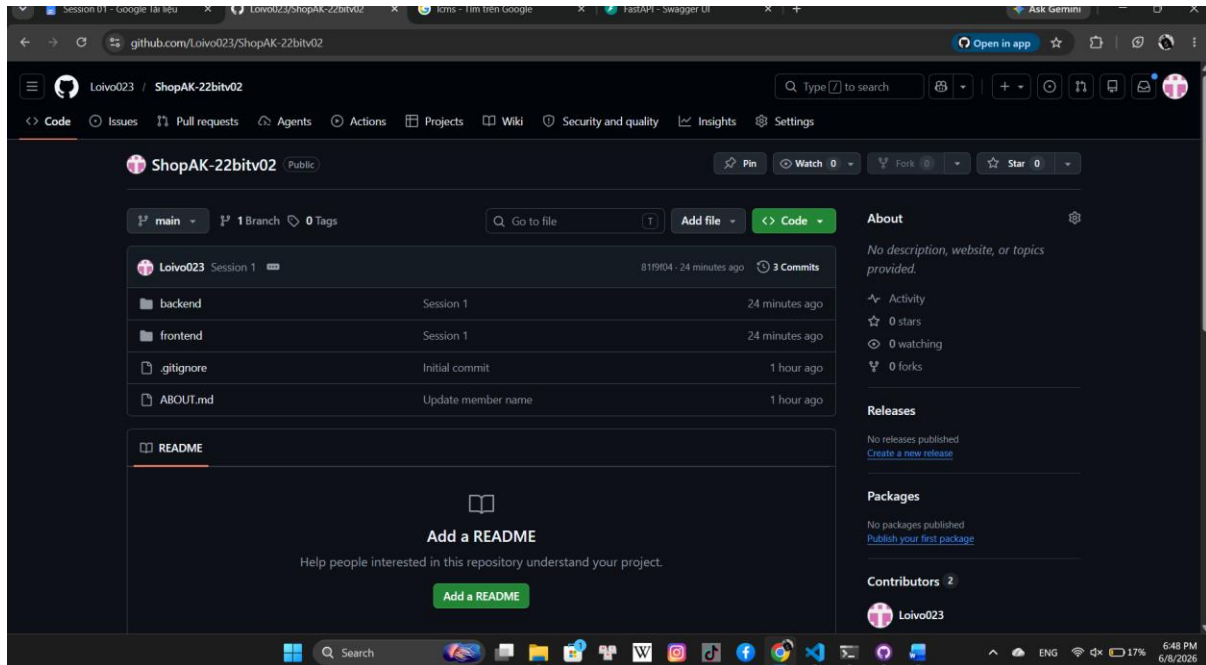
Period 1 – Setting up the Development Environment

1.1. Install Tools

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  SPELL CHECKER

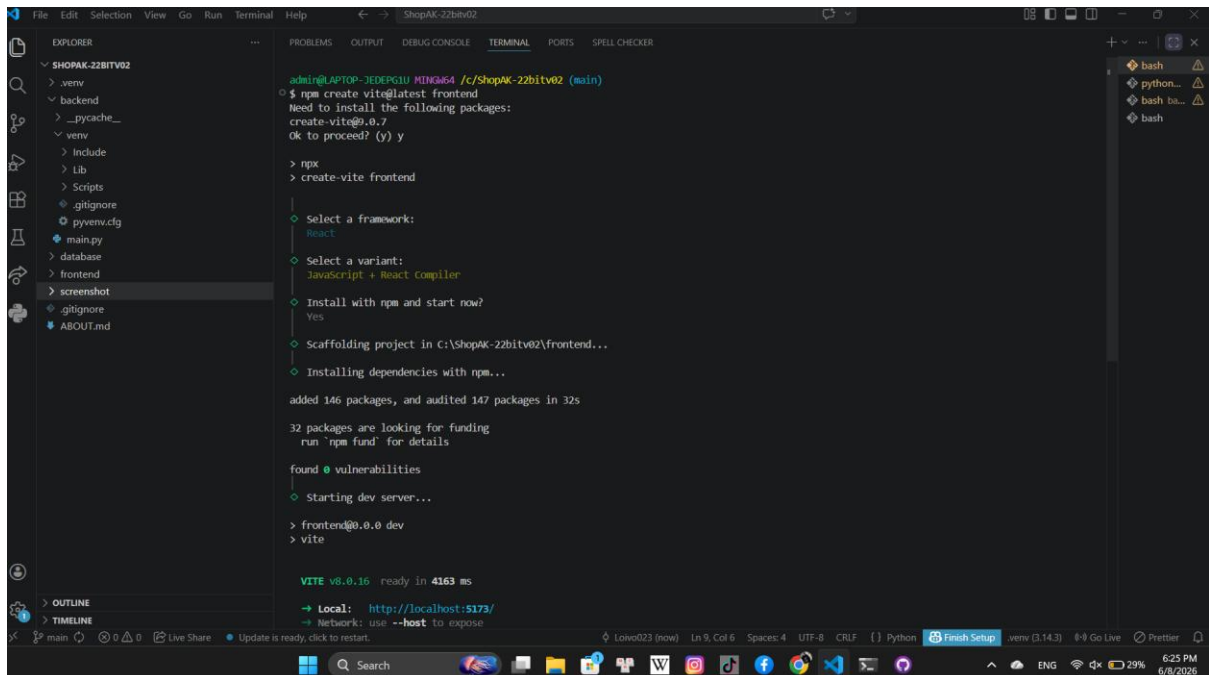
(.venv)
admin@LAPTOP-JEDEPG1U MINGW64 /c/ShopAK-22bitv02 (main)
$ node -v
(.venv)
admin@LAPTOP-JEDEPG1U MINGW64 /c/ShopAK-22bitv02 (main)
$ npm -v
11.11.0
(.venv)
admin@LAPTOP-JEDEPG1U MINGW64 /c/ShopAK-22bitv02 (main)
$ python --version
Python 3.14.3
(.venv)
admin@LAPTOP-JEDEPG1U MINGW64 /c/ShopAK-22bitv02 (main)
$ git --version
git version 2.52.0.windows.1
(.venv)
admin@LAPTOP-JEDEPG1U MINGW64 /c/ShopAK-22bitv02 (main)
$
```

1.2. GitHub Account & Repository



Period 2 – Creating the ReactJS Project

2.1. Create The frontend project



The screenshot shows a terminal window with the following commands and output:

```
admin@LAPTOP-JEDRGIU MINGW64 /c/ShopAK-22bitv02 (main)
$ npm create vite@latest frontend
Need to install the following packages:
create-vite@0.7
Ok to proceed? (y) y

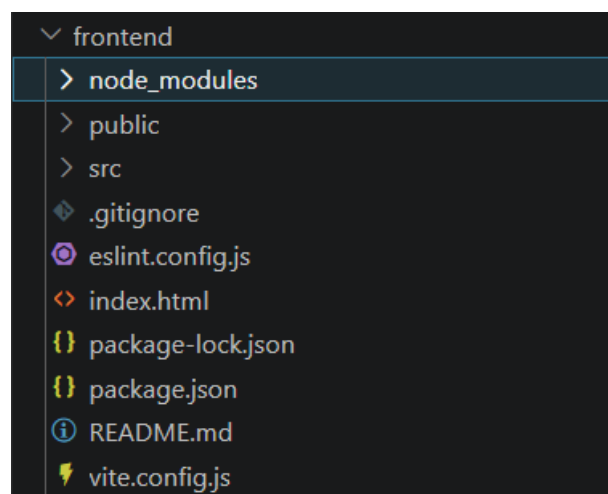
> npx
> create-vite frontend

|
| Select a framework:
|   React
|
| Select a variant:
|   JavaScript + React Compiler
|
| Install with npm and start now?
|   Yes
|
| Scaffolding project in c:\ShopAK-22bitv02\frontend...
|
| Installing dependencies with npm...
|
added 146 packages, and audited 147 packages in 32s
32 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
|
| Starting dev server...
|
> frontend@0.0.0 dev
> vite

VITE v0.0.16 ready in 4163 ms
→ Local: http://localhost:5173/
→ Network: use --host to expose
```

2.2. Vite Project Structure



2.3. Run the project

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS SPELL CHECKER

$ npm install
(.venv)
admin@LAPTOP-JEDEPG1U MINGW64 /c/ShopAK-22bitv02 (main)
$ cd frontend
(.venv)
admin@LAPTOP-JEDEPG1U MINGW64 /c/ShopAK-22bitv02/frontend (main)
$ npm run dev

> frontend@0.0.0 dev
> vite

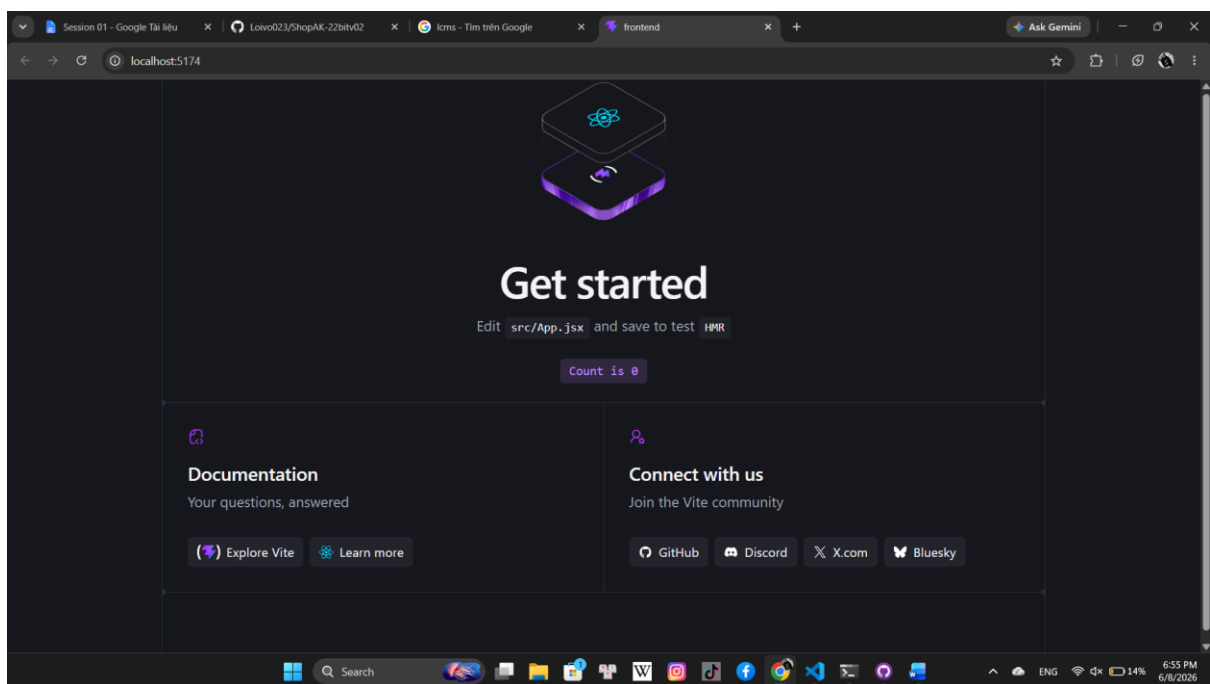
Port 5173 is in use, trying another one...

VITE v8.0.16 ready in 789 ms

→ Local: http://localhost:5174/
→ Network: use --host to expose
→ press h + enter to show help

Update is ready, click to restart.  Loivo023 (12 minutes ago) Ln 9, Col 6 Spaces: 4 UTF-8 CRLF Python Finish Setup .venv (3.14.3) Go L
```

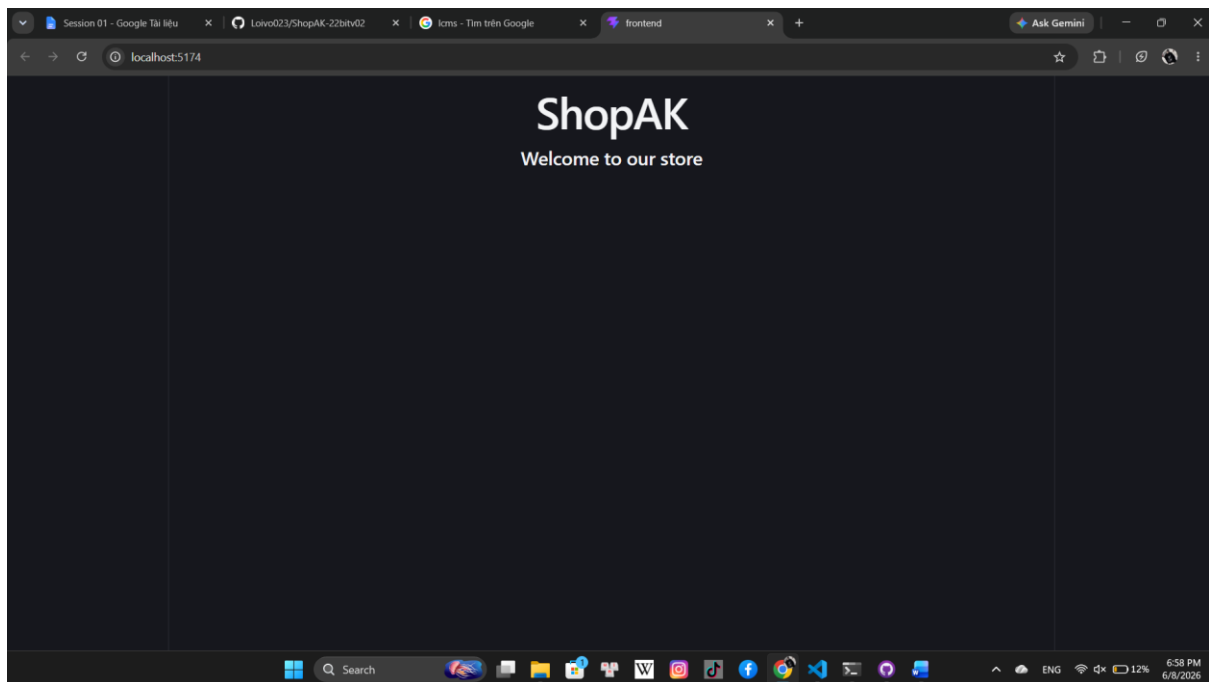
Results



2.4. First Hands-on Exercise

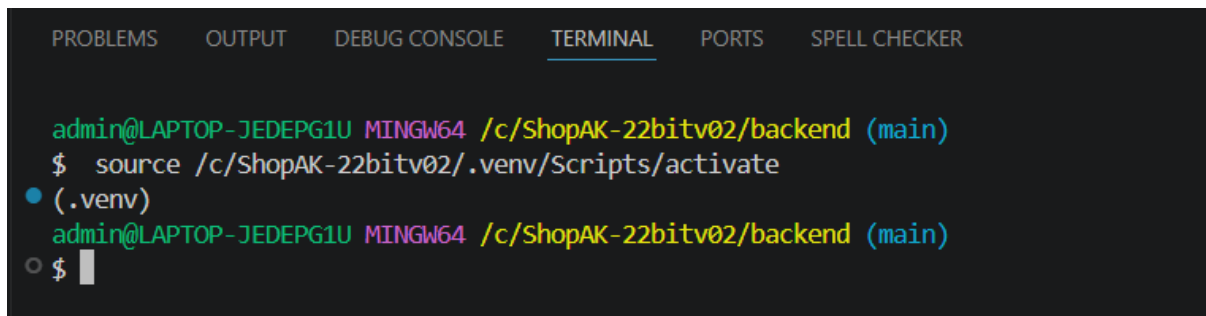
```
main.py App.jsx M Extension: Python
frontend > src > App.jsx > App
1 import { useState } from 'react'
2 import reactLogo from './assets/react.svg'
3 import viteLogo from './assets/vite.svg'
4 import heroImg from './assets/hero.png'
5 import './App.css'
6
7 function App() {
8   return (
9     <>
10    <h1>ShopAK</h1>
11    <h2>Welcome to our store</h2>
12    </>
13  )
14 }
15
16 export default App
17
```

Results



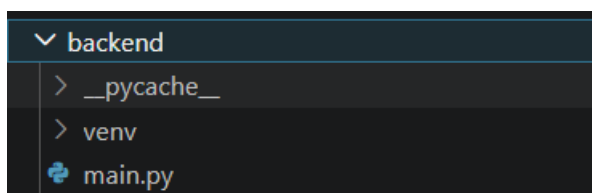
Period 3 – Creating the FastAPI Backend

3.1. Create the backend project

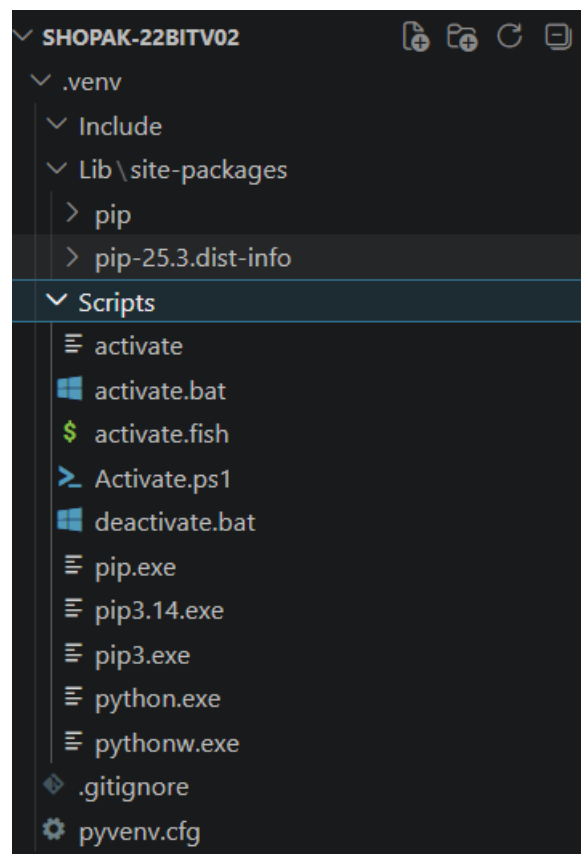


```
admin@LAPTOP-JEDEPG1U MINGW64 /c/ShopAK-22bitv02/backend (main)
$ source /c/ShopAK-22bitv02/.venv/Scripts/activate
• (.venv)
admin@LAPTOP-JEDEPG1U MINGW64 /c/ShopAK-22bitv02/backend (main)
$
```

Results



3.2. Create a virtual environment



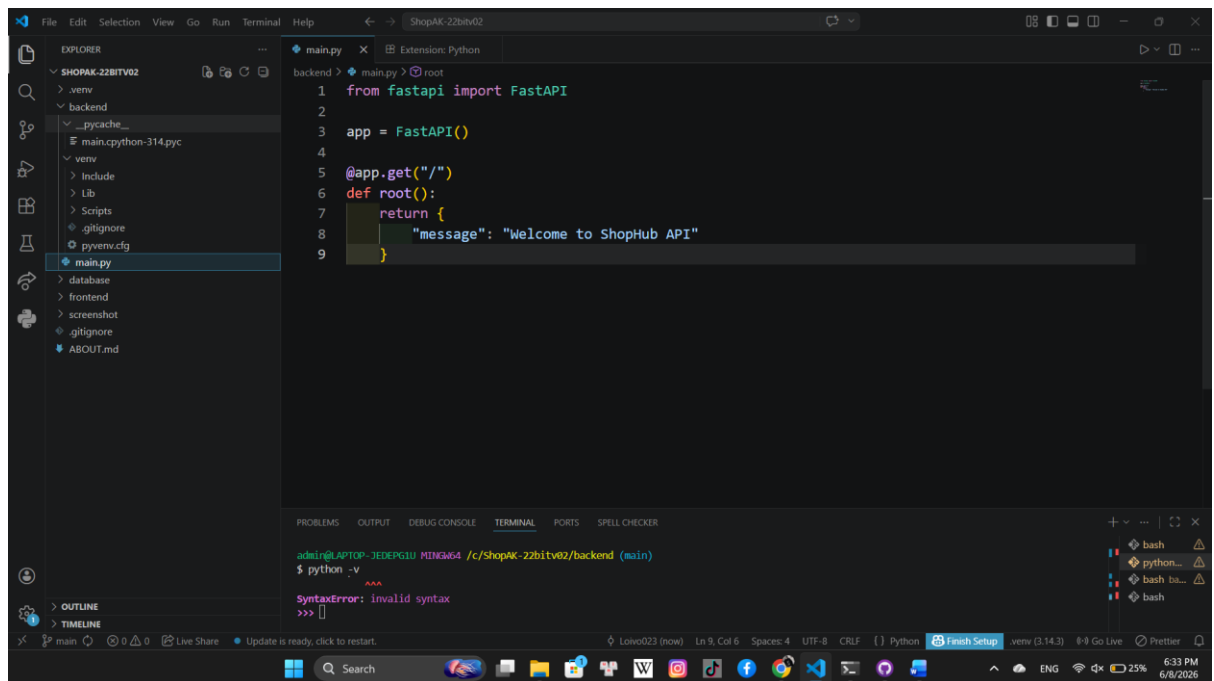
3.3. Install FastAPI and Uvicorn

```
C:\Windows\System32\cmd.exe
(venv) C:\ShopAK-22bitv02\backend>pythonpip install fastapi[standard]
'pythonpip' is not recognized as an internal or external command,
operable program or batch file.

(venv) C:\ShopAK-22bitv02\backend>pip install fastapi[standard]
Collecting fastapi[standard]
  Using cached fastapi-0.136.3-py3-none-any.whl.metadata (27 kB)
Collecting starlette>=0.46.0 (from fastapi[standard])
  Using cached starlette-1.2.1-py3-none-any.whl.metadata (6.3 kB)
Collecting pydantic>=2.9.0 (from fastapi[standard])
  Using cached pydantic-2.13.4-py3-none-any.whl.metadata (109 kB)
Collecting typing-extensions>=4.8.0 (from fastapi[standard])
  Using cached typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Collecting typing-inspection>=0.4.2 (from fastapi[standard])
  Using cached typing_inspection-0.4.2-py3-none-any.whl.metadata (2.6 kB)
Collecting annotated-doc>=0.0.2 (from fastapi[standard])
  Using cached annotated_doc-0.0.4-py3-none-any.whl.metadata (6.6 kB)
Collecting fastapi-cli>=0.0.8 (from fastapi-cli[standard]>=0.0.8; extra == "standard"->fastapi[standard])
  Using cached fastapi_cli-0.0.24-py3-none-any.whl.metadata (6.4 kB)
Collecting fastar>=0.9.0 (from fastapi[standard])
  Using cached fastar-0.11.0-cp314-cp314-win_amd64.whl.metadata (4.1 kB)
Collecting httpx<1.0.0,>=0.23.0 (from fastapi[standard])
  Using cached httpx-0.28.1-py3-none-any.whl.metadata (7.1 kB)
Collecting jinja2>=3.1.5 (from fastapi[standard])
  Using cached jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting python-multipart>=0.0.18 (from fastapi[standard])
  Using cached python_multipart-0.0.32-py3-none-any.whl.metadata (2.1 kB)
Collecting email-validator>=2.0.0 (from fastapi[standard])
  Using cached email_validator-2.3.0-py3-none-any.whl.metadata (26 kB)
Collecting uvicorn>=0.12.0 (from uvicorn[standard]>=0.12.0; extra == "standard"->fastapi[standard])
```

```
(venv) C:\ShopAK-22bitv02\backend>pip list
Package           Version
-----
annotated-doc     0.0.4
annotated-types   0.7.0
anyio             4.13.0
certifi           2026.5.20
click            8.4.1
colorama          0.4.6
detect-installer  0.1.0
dnspython         2.8.0
email-validator   2.3.0
fastapi           0.136.3
fastapi-cli       0.0.24
fastapi-cloud-cli 0.19.0
fastar            0.11.0
h11              0.16.0
httpcore          1.0.9
httpx             0.28.1
idna              3.18
Jinja2            3.1.6
markdown-it-py    4.2.0
MarkupSafe        3.0.3
mdurl             0.1.2
pip              25.3
pydantic          2.13.4
pydantic_core     2.46.4
pydantic-extra-types 2.11.1
pydantic-settings 2.14.1
Pygments          2.20.0
python-dotenv     1.2.2
python-multipart  0.0.32
PyYAML            6.0.3
rich              15.0.0
rich-toolkit      0.20.1
rignore           0.7.6
sentry-sdk        2.61.1
shellingham       1.5.4
starlette         1.2.1
typer            0.26.7
typing_extensions 4.15.0
typing-inspection 0.4.2
urllib3           2.7.0
uvicorn           0.49.0
watchfiles        1.2.0
websockets        16.0
```

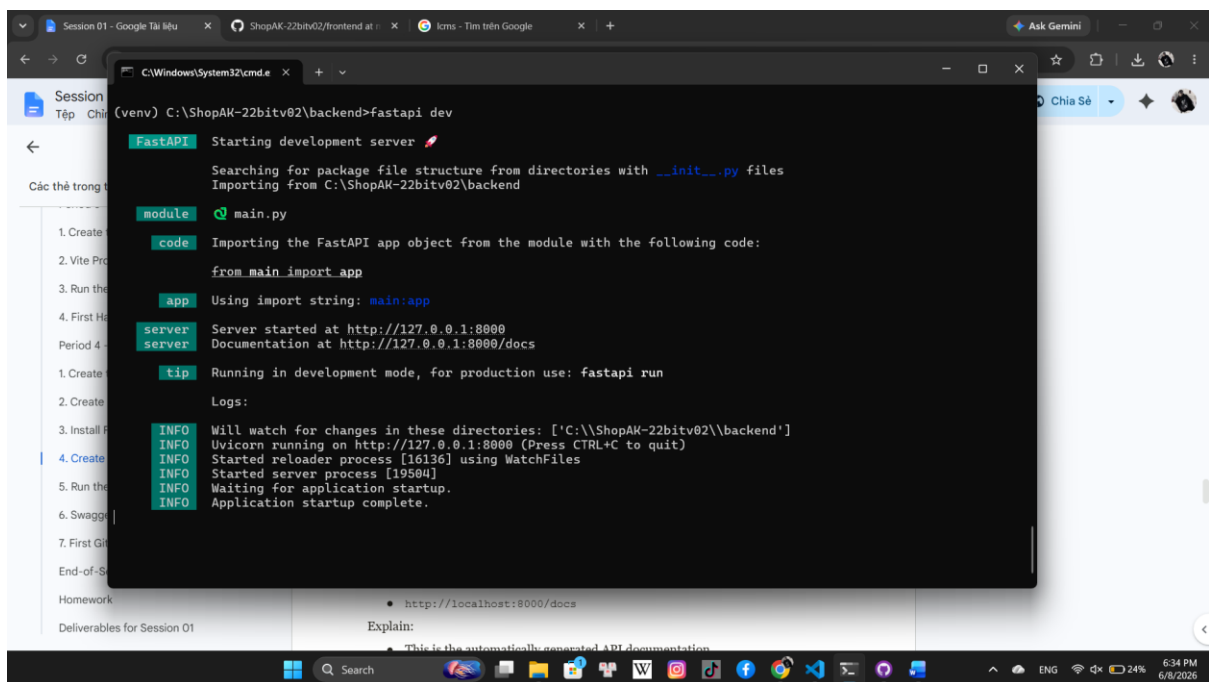
3.4. Create main.py



```
1 from fastapi import FastAPI
2
3 app = FastAPI()
4
5 @app.get("/")
6 def root():
7     return {
8         "message": "Welcome to ShopHub API"
9     }
```

```
admin@LAPTOP-JEDEPGUJ MING3654 /c/ShopAK-22bitv02/backend (main)
$ python -v
SyntaxError: invalid syntax
>>> |
```

3.5. Run the server



```
(venv) C:\ShopAK-22bitv02\backend>fastapi dev

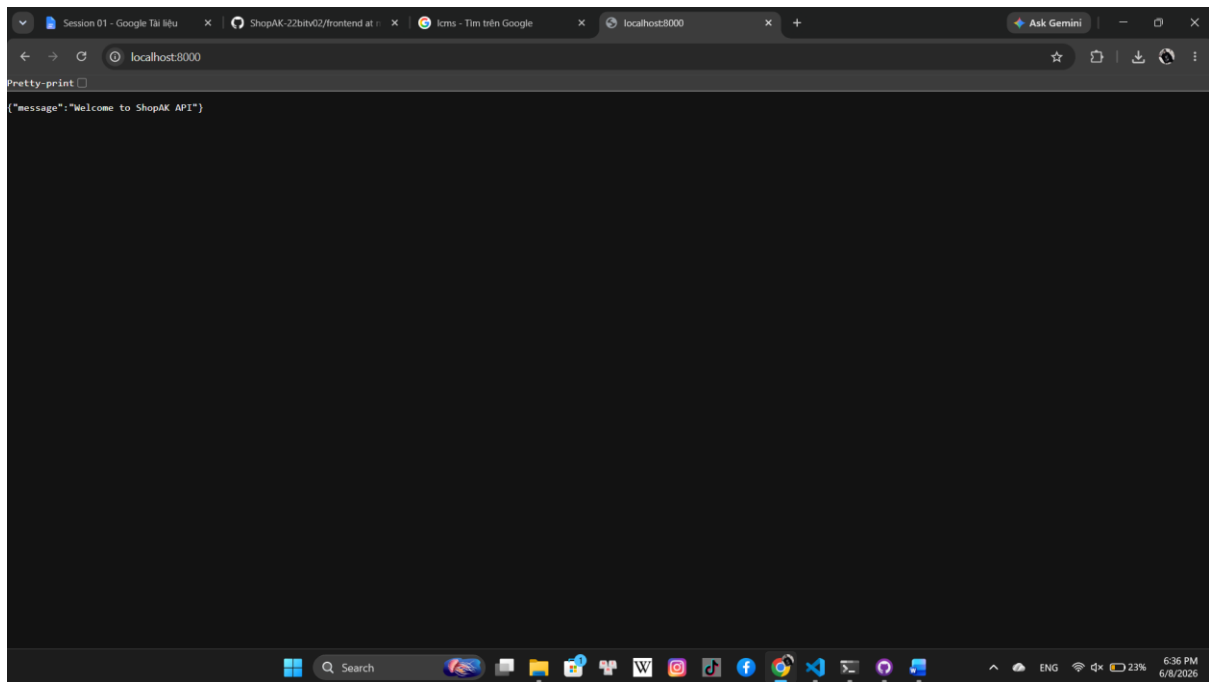
FastAPI Starting development server

Searching for package file structure from directories with __init__.py files
Importing from C:\ShopAK-22bitv02\backend

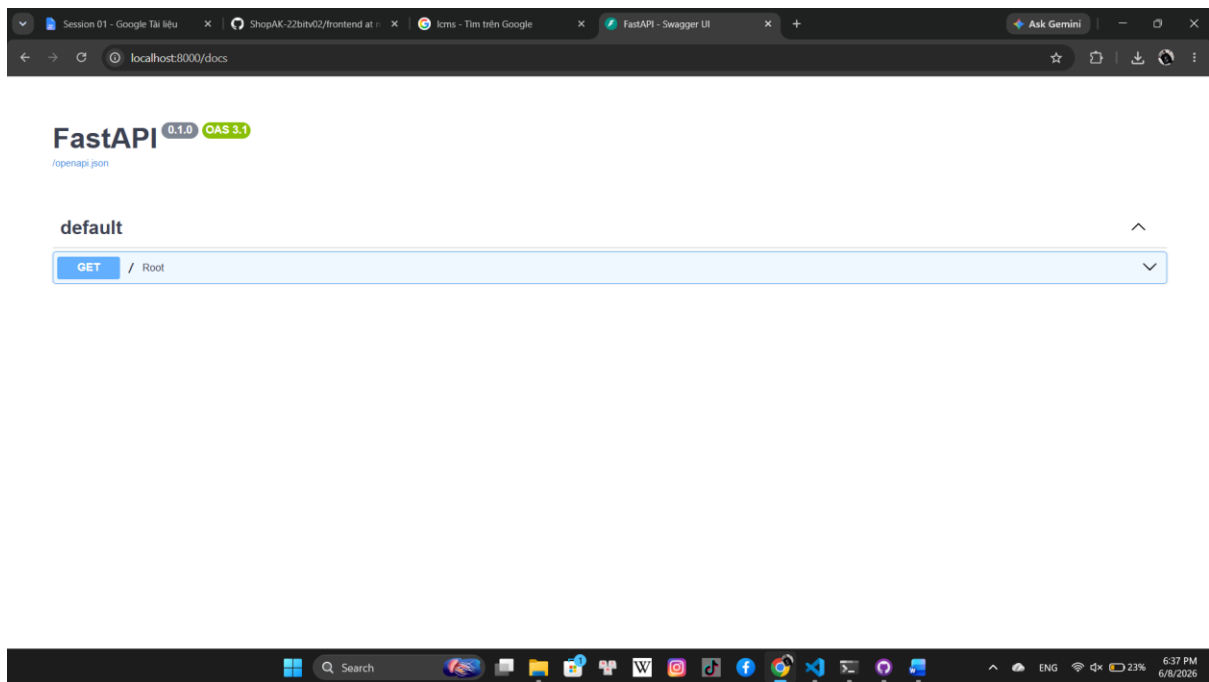
module main.py
code Importing the FastAPI app object from the module with the following code:
from main import app
app Using import string: main:app
server Server started at http://127.0.0.1:8000
server Documentation at http://127.0.0.1:8000/docs
tip Running in development mode, for production use: fastapi run

Logs:
INFO Will watch for changes in these directories: ['C:\\ShopAK-22bitv02\\backend']
INFO Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO Started reloader process [16136] using WatchFiles
INFO Started server process [19504]
INFO Waiting for application startup.
INFO Application startup complete.
```

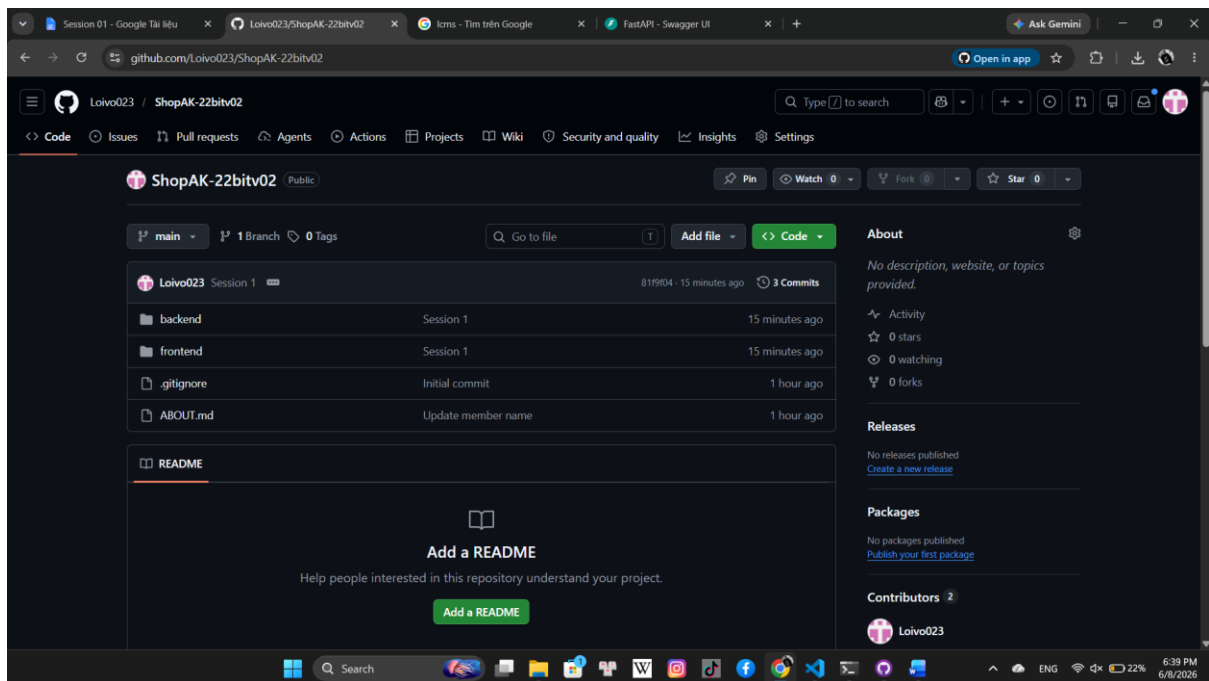
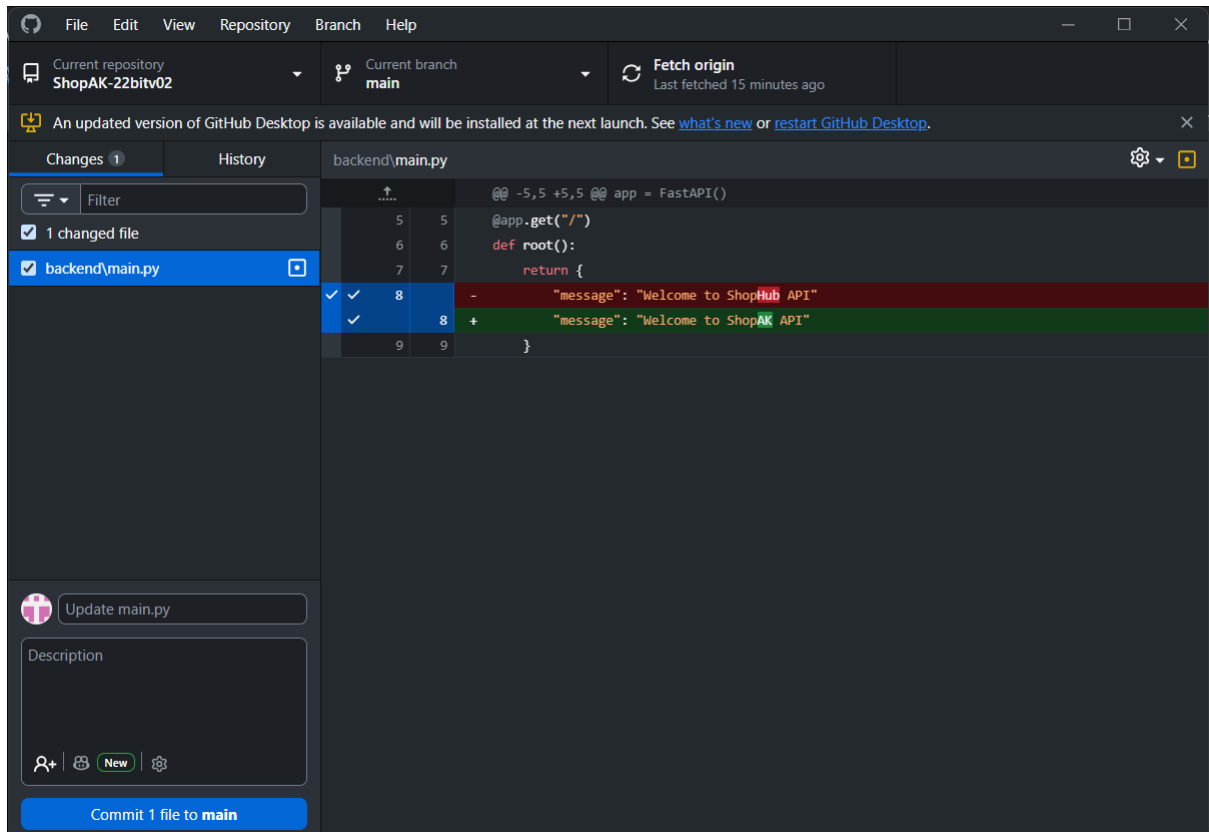
Results



3.6. Swagger UI

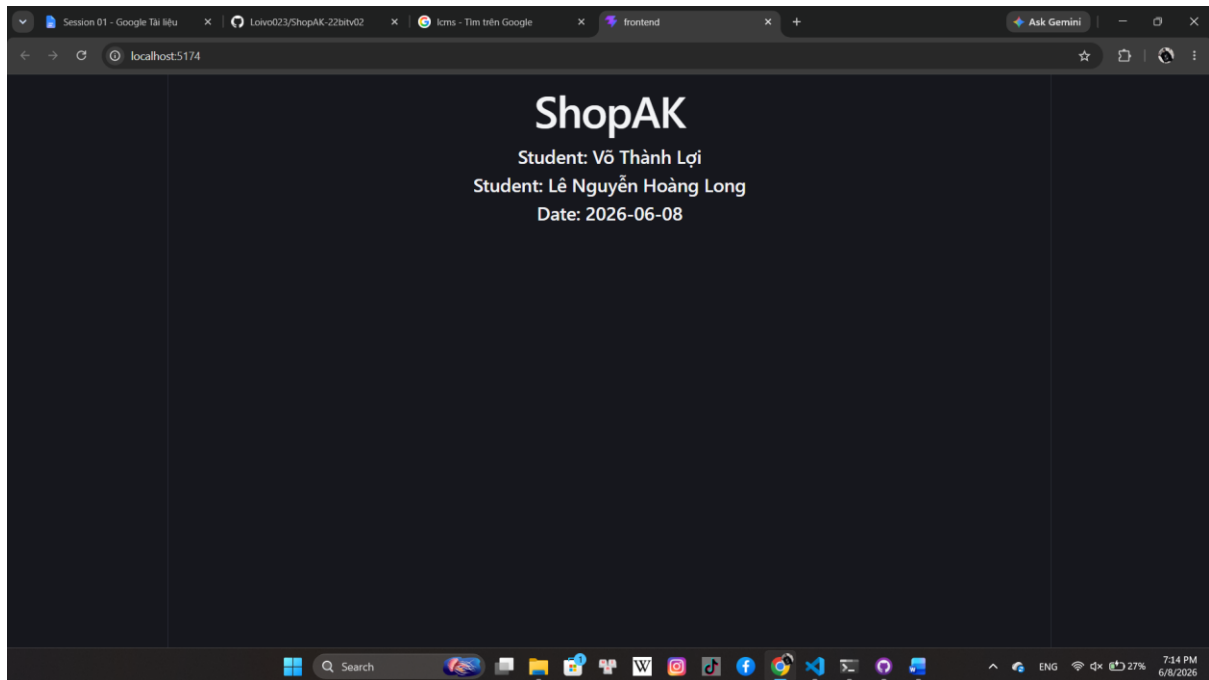


3.7. First git commit

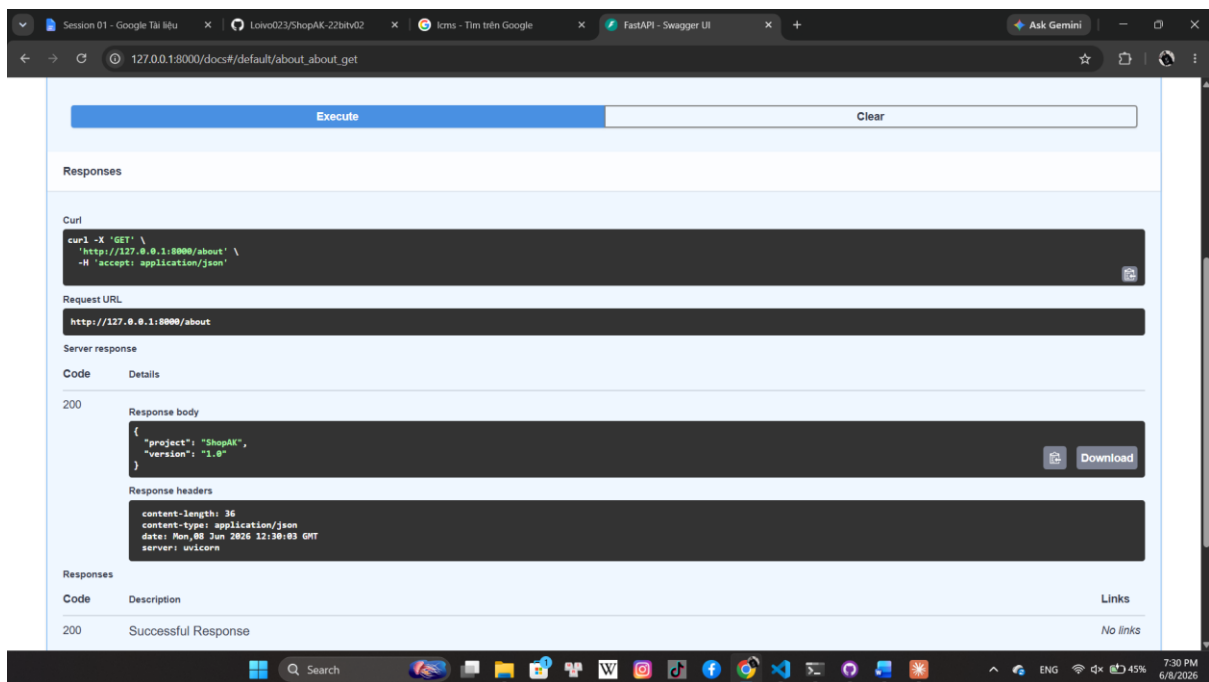


End-of-Session Lab

Frontend



Backend



HOME WORK

Exercise 2:

The screenshot displays a web browser window with multiple tabs. The active tab is 'FastAPI - Swagger UI', showing the Swagger interface for a REST API. The URL bar indicates the endpoint `127.0.0.1:8000/docs#/default/get_all_product_products_get`.

The 'Responses' section is expanded, showing details for a `200` status code:

- Curl:**

```
curl -X 'GET' \
  'http://127.0.0.1:8000/products' \
  -H 'accept: application/json'
```
- Request URL:** `http://127.0.0.1:8000/products`
- Server response:**
- Code:** `200`
- Details:**
- Response body:**

```
{
  "id": 1,
  "name": "Laptop"
},
{
  "id": 2,
  "name": "Mouse"
}
```
- Response headers:**

```
content-length: 59
content-type: application/json
date: Mon, 08 Jun 2026 12:34:46 GMT
server: uvicorn
```
- Responses table:**

Code	Description	Links
200	Successful Response	No links

The Windows taskbar at the bottom shows the system clock as 7:34 PM on 6/8/2026, with a battery level of 48%.